



Algorithms: Greedy Method

An Activity-Selection Problem

Greedy Algorithms: Principles

- A *greedy algorithm* always makes the choice that looks best at the moment.
- A greedy algorithm works in phases.
At each phase:
 - You take the **best you can get right now**, without regard for future consequences.
 - You hope that by choosing a local optimum at each step, you will end up at a global optimum.
 - For some problems, it works.



An Activity Selection Problem

- Input: A set of activities $S = \{a_1, \dots, a_n\}$
 - Each activity a_i has a start time s_i and a finish time f_i , where $0 \leq s_i < f_i < \infty$
 - If selected, activity a_i takes place during the half-open time interval $[s_i, f_i)$
- Two activities are **compatible** if and only if their intervals do not overlap
- Output: a maximum-size subset of mutually compatible activities

The Activity Selection Problem

- Here are a set of start and finish times

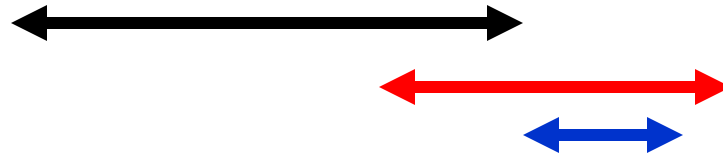
i	1	2	3	4	5	6	7	8	9	10	11
s_i	1	3	0	5	3	5	6	8	8	2	12
f_i	4	5	6	7	8	9	10	11	12	13	14

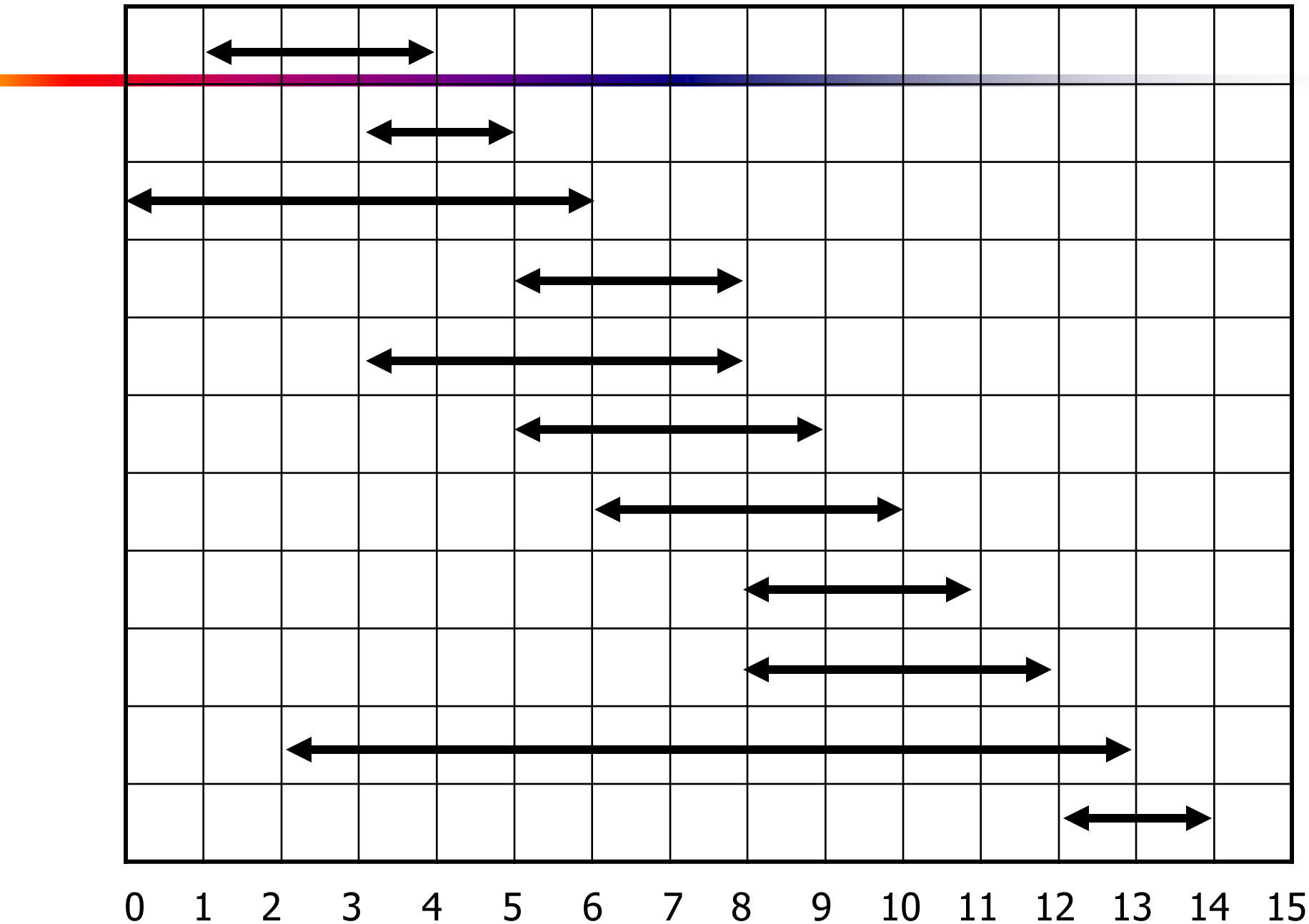
- What is the maximum number of activities that can be completed?
 - $\{a_3, a_9, a_{11}\}$ can be completed
 - But so can $\{a_1, a_4, a_8, a_{11}\}$ which is a larger set
 - But it is not unique, consider $\{a_2, a_4, a_9, a_{11}\}$

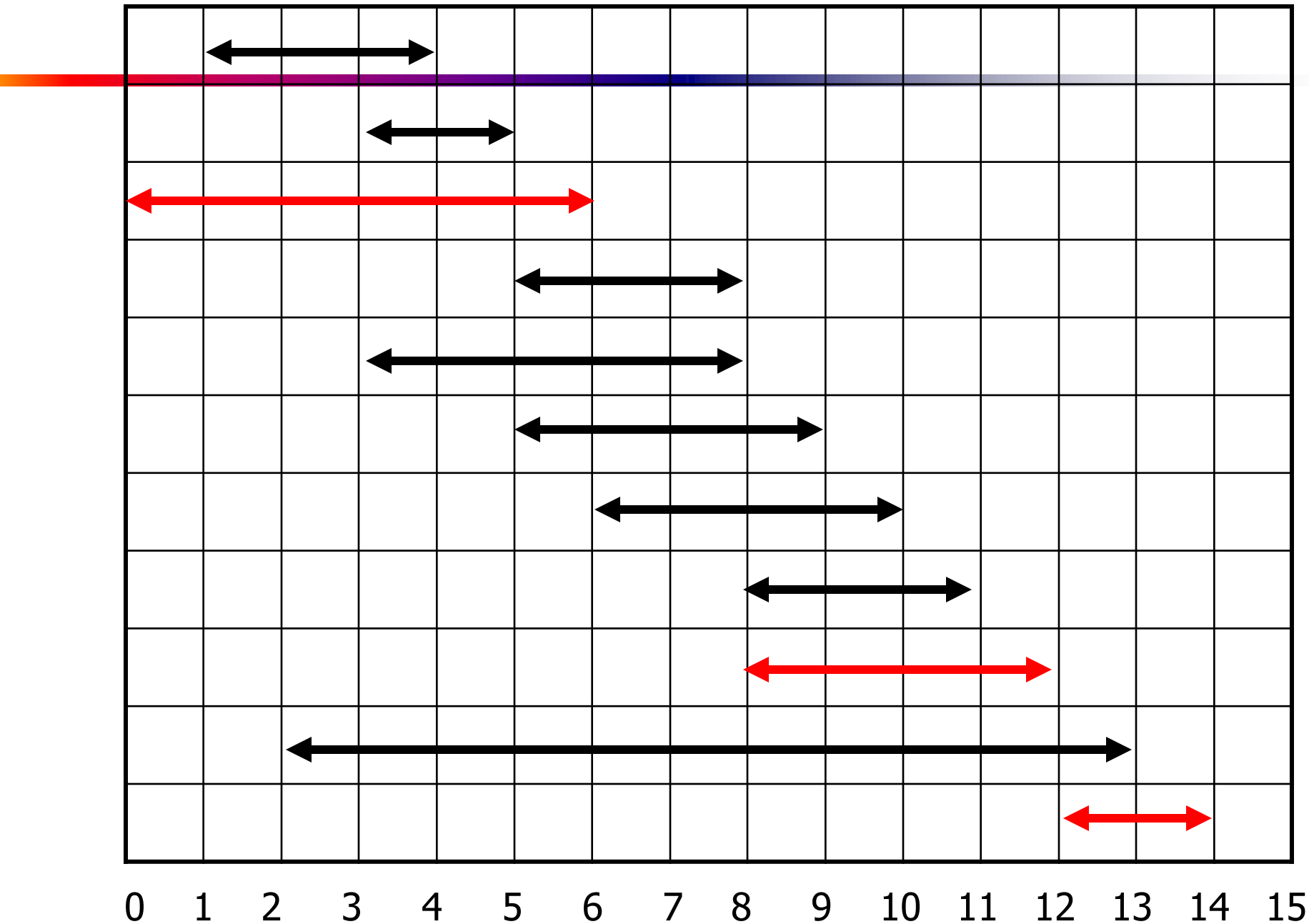
Interval Representation

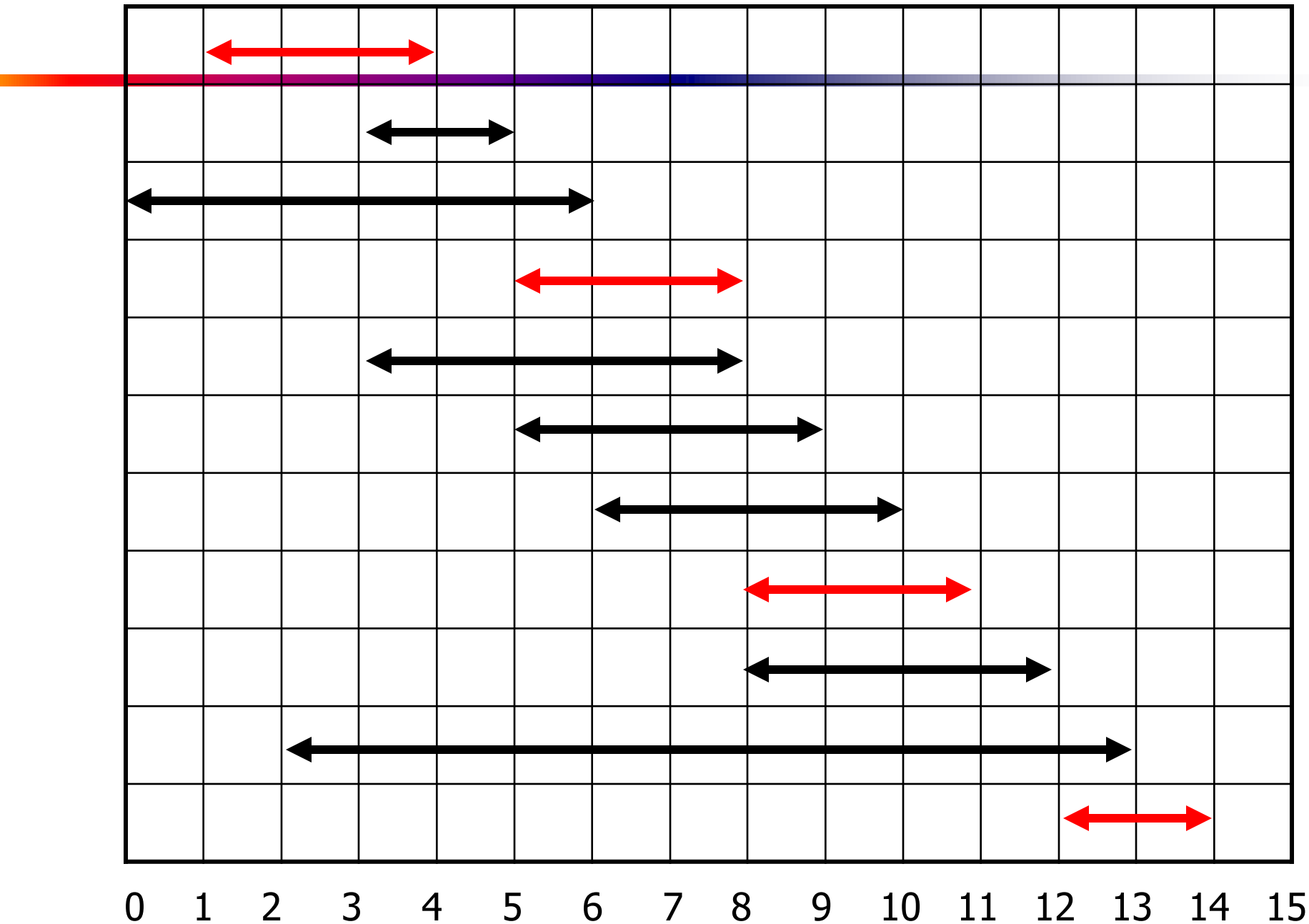
- Here are a set of start and finish times

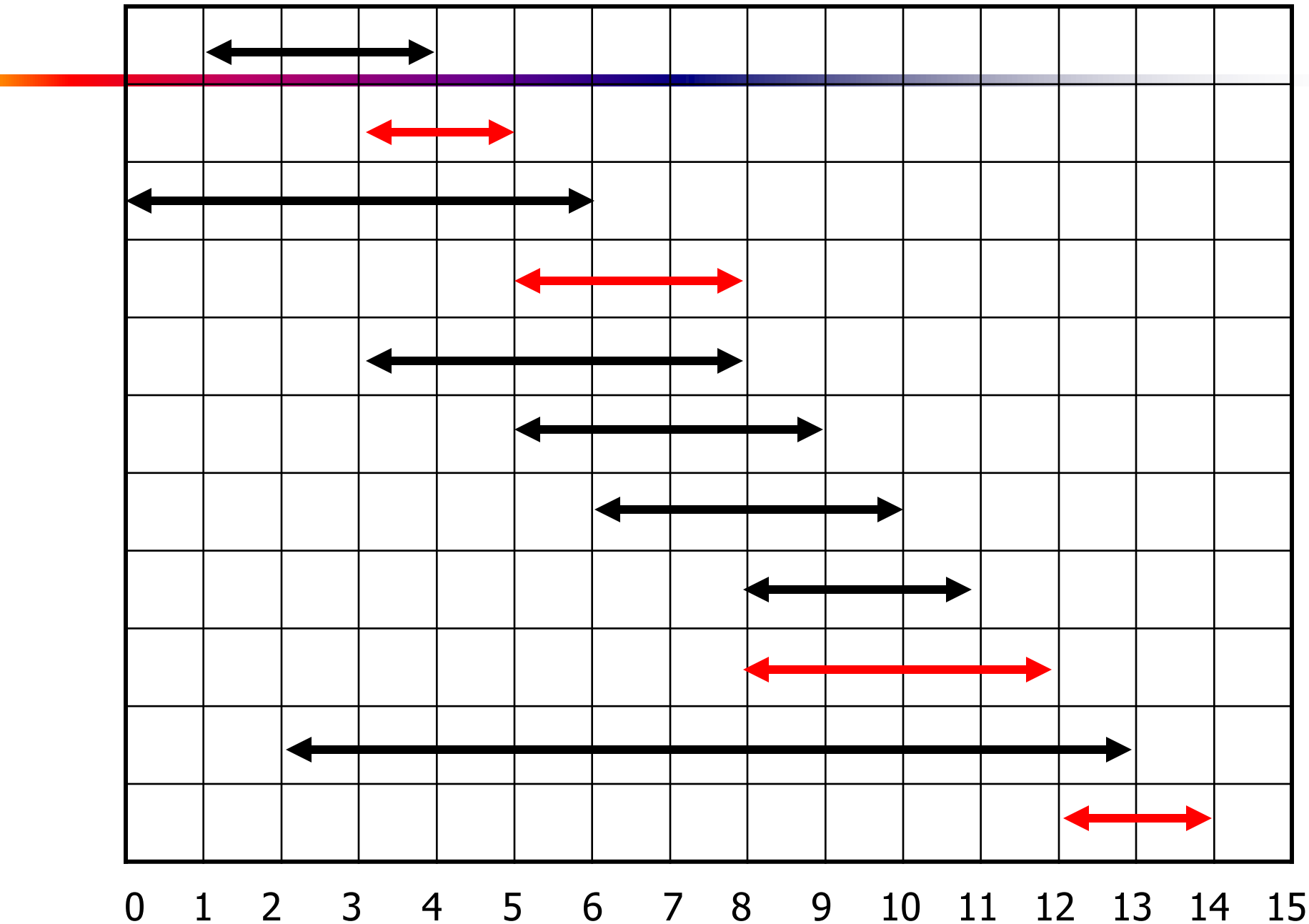
i	1	2	3	4	5	6	7	8	9	10	11
s_i	1	3	0	5	3	5	6	8	8	2	12
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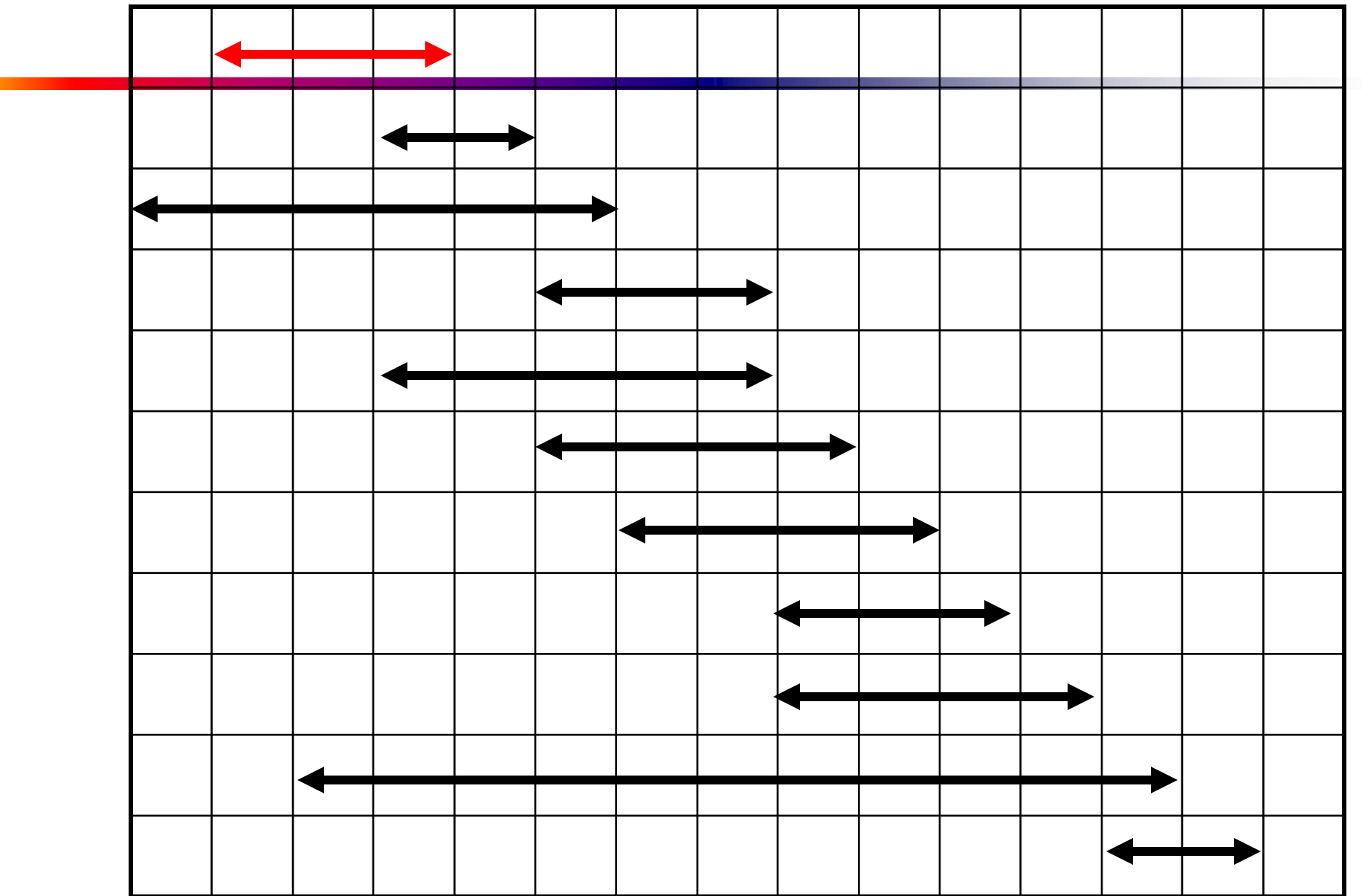


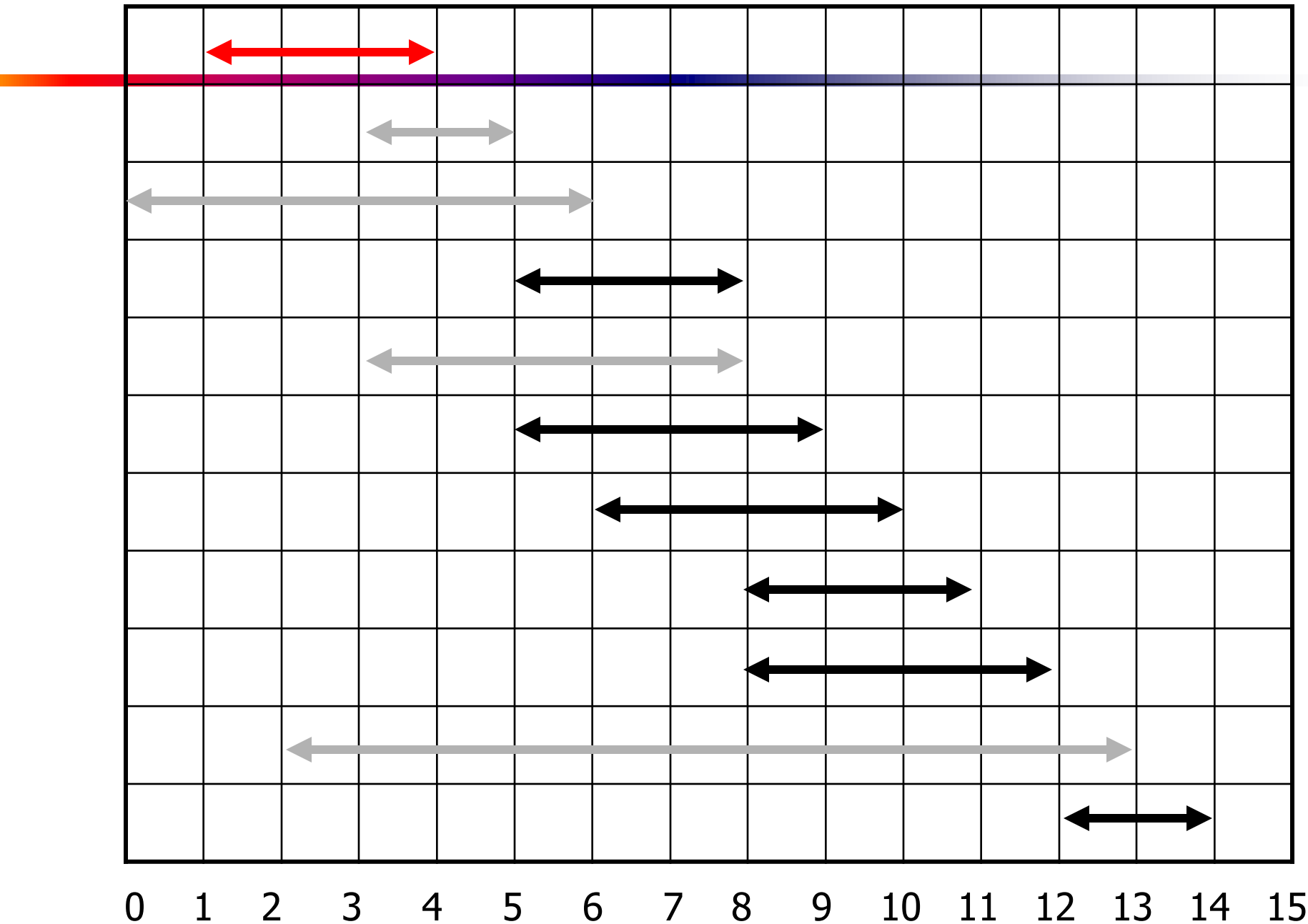


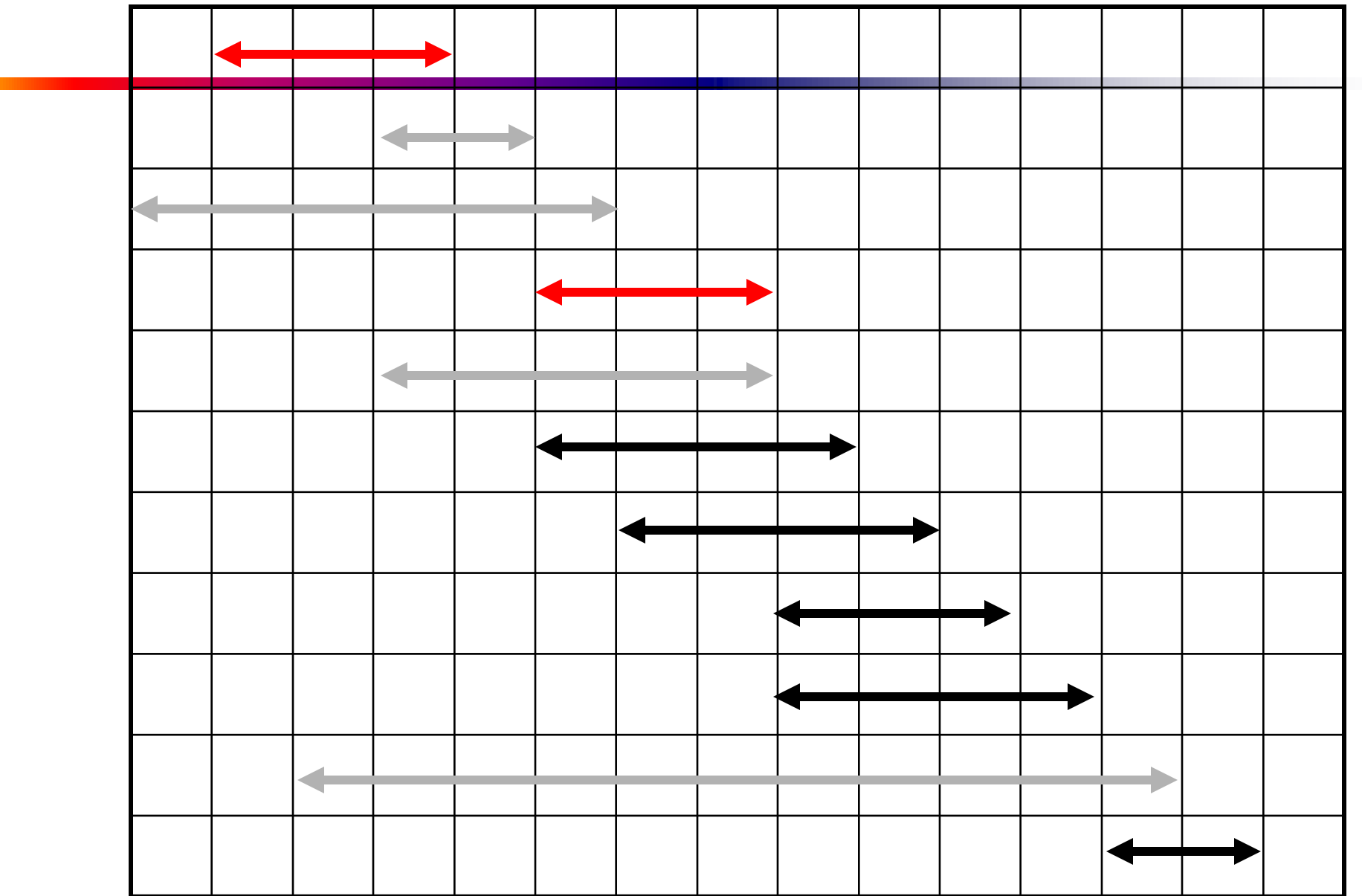


Early Finish Greedy

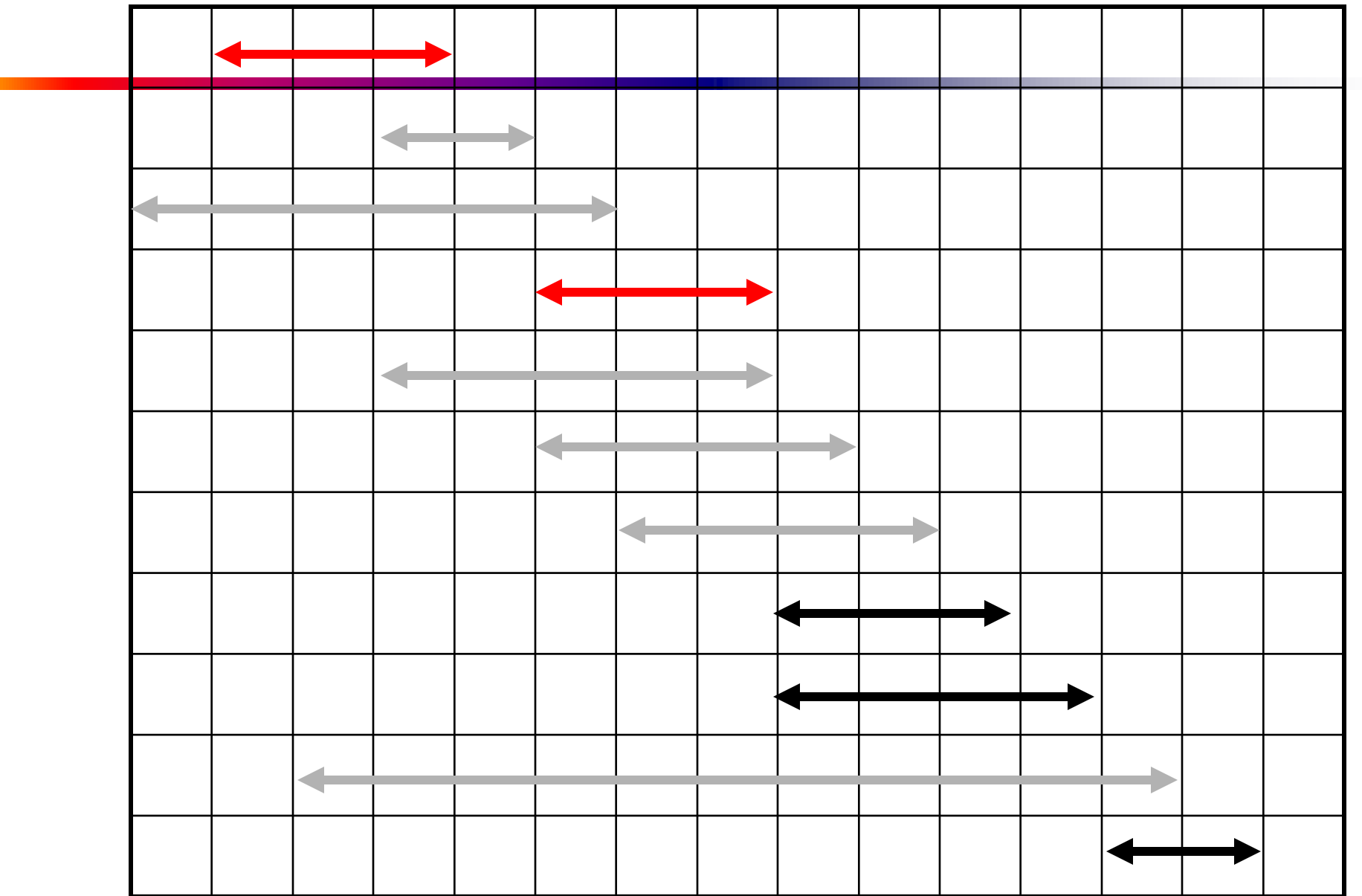
- Select the activity with the earliest finish
- Eliminate the activities that could not be scheduled
- Repeat!



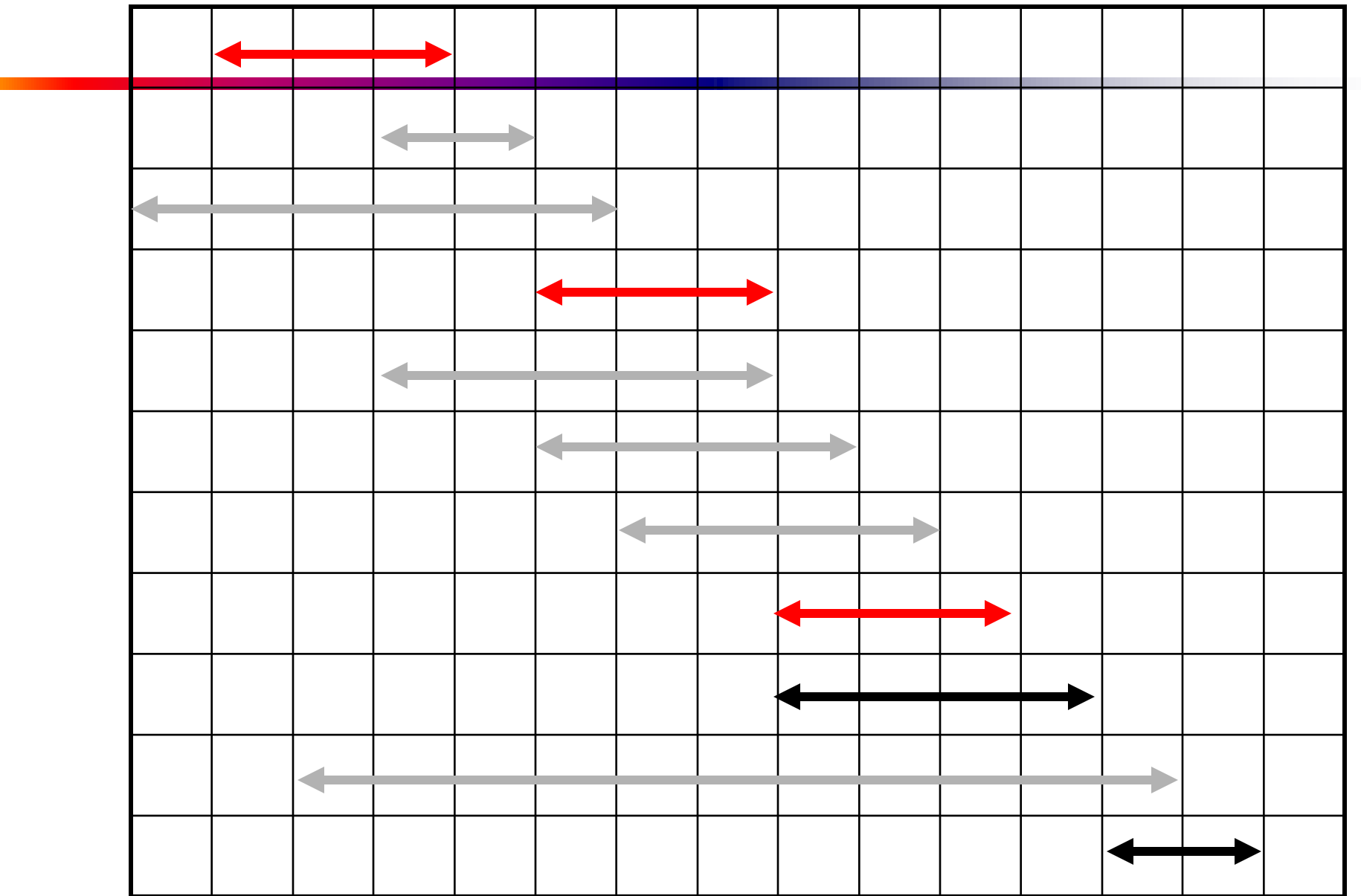




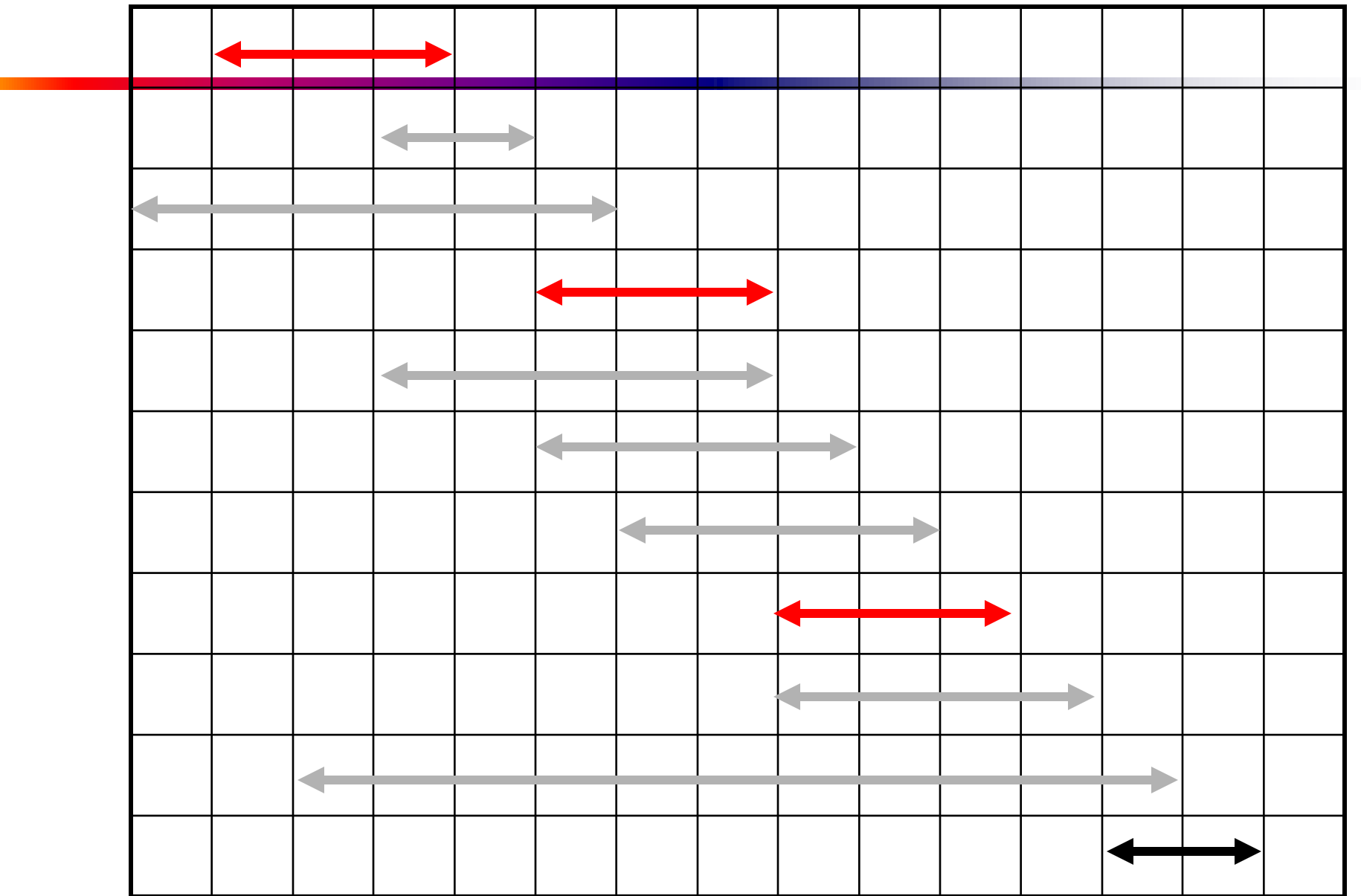
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15



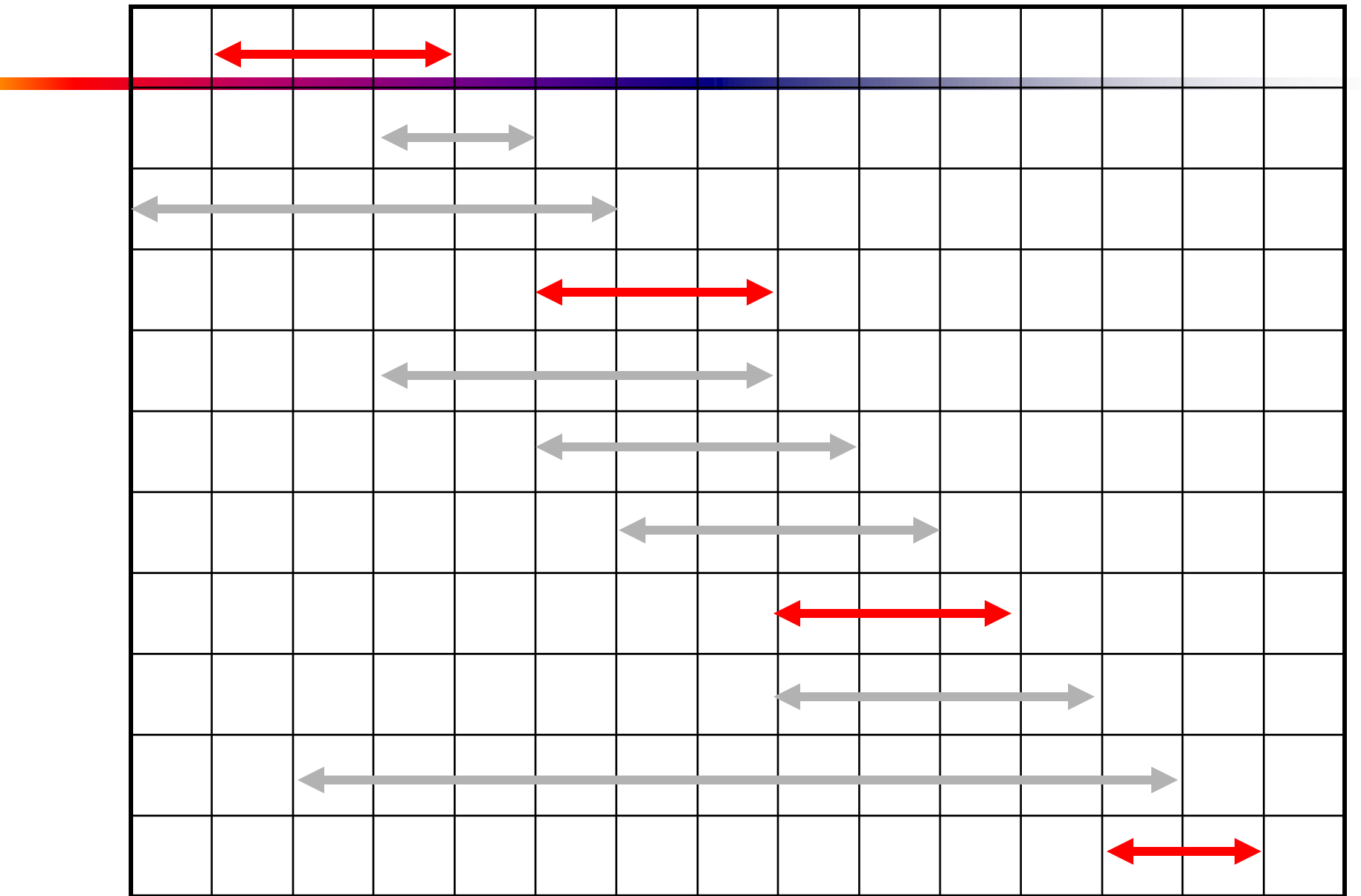
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15



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Assuming activities are sorted by finish time

GREEDY-ACTIVITY-SELECTOR(s, f)

```
1   $n \leftarrow \text{length}[s]$ 
2   $A \leftarrow \{a_1\}$ 
3   $i \leftarrow 1$ 
4  for  $m \leftarrow 2$  to  $n$ 
5      do if  $s_m \geq f_i$ 
6          then  $A \leftarrow A \cup \{a_m\}$ 
7               $i \leftarrow m$ 
8  return  $A$ 
```